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RIVET-H3: Four-Channel SHA-256, HMAC-SHA-256 and SHA-512 RTL IP

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:18 UTC+8 Beijing Time)

Abstract

RIVET-H3 is a synthesizable cryptographic hashing subsystem with four independently tagged channels, streaming message and key inputs, portable context save/restore, and backpressure-safe digest delivery. It preserves the six original RIVET-H2 modules unchanged and adds an iterative SHA-512 datapath to each channel. Independent Icarus verification completed six seeded behavior configurations with 3,504 complete digest comparisons and zero mismatches, plus 244 compatibility comparisons and 25 cycle-measurement workloads. For 1,024-byte messages, measured aggregate throughput increased from 0.4171 to 1.2876 bytes/cycle for SHA-256, from 0.4060 to 1.2611 for HMAC-SHA-256, and from 0.5002 to 1.4644 for SHA-512 when moving from one to four active channels. These are simulation-cycle results, not physical frequency or implementation measurements.

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The top module, rivet_h3_top, replicates four channel engines rather than time-sharing one compressor. Each g_channel[n].u_channel contains an unchanged rivet_h2_top for SHA-256/HMAC and a rivet_h3_sha512 engine. Only the selected algorithm receives message traffic or exposes responses. This arrangement preserves parent behavior and isolates channel progress at the cost of replicated storage and arithmetic. Shared tagged input ports limit injection bandwidth; separate digest and context arbiters serialize responses. The following blocks describe concurrent responsibilities, not a serial processing pipeline. Sources: E001–E013.

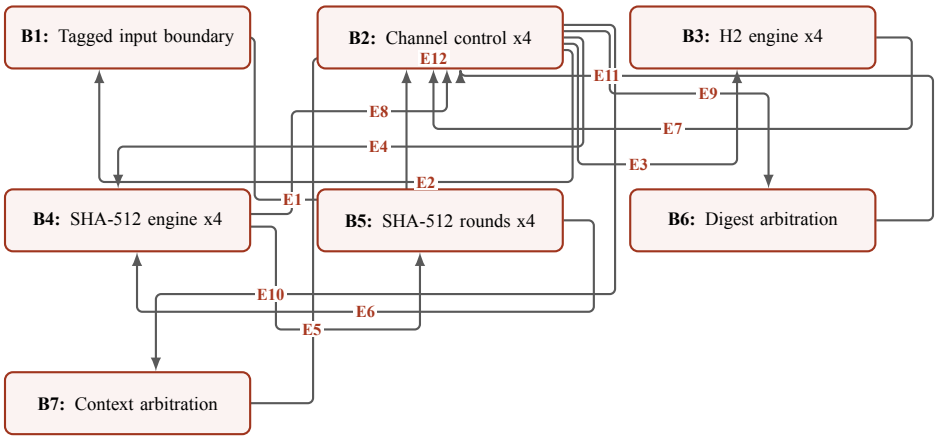


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Tagged input boundary	Routes message, key and command handshakes to the addressed channel; returns readiness from that channel only.
B2: Channel control x4	Owns algorithm selection, command priority, fault isolation, context-envelope validation and inactive-engine cleanup.
B3: H2 engine x4	Preserves SHA-256, long-key HMAC normalization, cached inner/outer seeds and original context processing.
B4: SHA-512 engine x4	Owns beat staging, 128-byte block collection, 128-bit byte count, chaining state, padding and snapshots.
B5: SHA-512 rounds x4	Performs 80 iterative 64-bit rounds with a sixteen-word sliding schedule and separate feed-forward completion.
B6: Digest arbitration	Locks one pending tagged digest until retirement and advances an independent round-robin pointer.
B7: Context arbitration	Independently locks one tagged 2304-bit snapshot until retirement.

Directed relationship	Meaning
E1: B1 to B2	Selected message, key and command transfers
E2: B2 to B1	Selected-channel readiness
E3: B2 to B3	SHA/HMAC traffic and control
E4: B2 to B4	SHA-512 traffic and control
E5: B4 to B5	Prepared block, chain and launch
E6: B5 to B4	Feed-forward state and registered completion
E7: B3 to B2	H2 digest, snapshot and status
E8: B4 to B2	SHA-512 digest, snapshot and status
E9: B2 to B6	Selected-algorithm pending digest
E10: B2 to B7	Enveloped pending snapshot
E11: B6 to B2	Digest retirement
E12: B7 to B2	Snapshot retirement

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge clock and asynchronous active-low reset. While reset is asserted, every public ready, valid, payload, tag, algorithm and error output is zero.
s_valid, s_ready, s_ channel, s_data, s_keep, s_last	Input except output s_ ready	1, 1, 2, 32, 4, 1 bits	Message stream. Transfer occurs only on a rising edge with valid and ready asserted. Channel indices are 0–3. Leading bytes start at data[31:24]; keep masks 1000, 1100, 1110 and 1111 encode one through four bytes.
k_valid, k_ready, k_ channel, k_data, k_keep, k_last	Input except output k_ ready	1, 1, 2, 32, 4, 1 bits	HMAC key stream using the same byte and mask convention. Enabled during KEY_LOAD; key completion produces no digest.

Signal / group	Direction	Width	Contract
cmd_valid, cmd_ready, cmd_channel, cmd_op, cmd_context	Input except output cmd_ ready	1, 1, 2, 3, 2304 bits	Commands: 0 SHA_INIT, 1 KEY_LOAD, 2 HMAC_ INIT, 3 SAVE, 4 RESTORE, 5 CANCEL, 6 SHA512_ INIT. Accepted opcode 7 faults the addressed channel. An accepted command suppresses same-channel stream transfers.
m_valid, m_ready, m_ channel, m_alg, m_ digest	Output except input m_ready	1, 1, 2, 2, 512 bits	Locked digest response. Algorithm codes are 0 SHA-256, 1 HMAC-SHA-256 and 2 SHA-512; 3 is reserved. SHA-256/HMAC occupy bits [255:0] with zero upper bits. Digest bytes follow published big-endian order.
ctx_valid, ctx_ready, ctx_channel, ctx_data	Output except input ctx_ ready	1, 1, 2, 2304 bits	Independent locked context response. The channel tag is outside the serialized state, enabling migration.
error	Output	4 bits	One sticky fault bit per channel. A faulty operation emits no digest; only CANCEL or reset recovers the channel.

Parameter	Default	Meaning / supported values
PARTIAL_BEATS	0	Legal integer values are exactly 0 and 1, both exercised on H3 and standalone H2. Value 0 requires full nonfinal beats and permits a zero-byte final beat only for an empty message/key. Value 1 permits leading partial nonfinal beats and zero-byte final terminators. Sparse masks remain invalid. Values -1 and 2 were explicitly rejected with PARAMETER ERROR and nonzero simulation exit. Channel count is fixed at four.

III Behavior and corner cases

A Collection and concurrency

Each active channel has one beat stage and one block buffer. A beat captured at E0 first drains at E1; four bytes finish at E4, with no replacement on E4. A crossing beat pauses across block compression without losing bytes. Command, key and message ports may transfer simultaneously to different channels. A pending response blocks further work on its channel, but other channels can continue to their own response storage. No second message enters a channel before its prior digest retires.

B Exact cryptographic arithmetic

SHA-256 and SHA-512 follow FIPS 180-4; HMAC-SHA-256 follows RFC 2104/RFC 4231, hashing keys longer than 64 bytes. SHA-512 uses eight 64-bit chaining words, standard IV/constants, 128-byte blocks and 80 rounds with modulo-2^64 addition. Its schedule is W[t]=sigma1(W[t-2])+W[t-7]+sigma0(W[t-15])+W[t-16], where sigma0 uses ROTR1/ROTR8/SHR7 and sigma1 uses ROTR19/ROTR61/SHR6. Large sigmas use rotations 28/34/39 and 14/18/41. Ch and Maj use the standard bitwise definitions.

C Padding and length limits

SHA-512 appends 0x80, zeros and the complete 128-bit big-endian bit length. Residual lengths 112−127 require an extra padding block; an exactly full final data block precedes a fresh padding block. The full 128-bit byte counter rejects prospective counts above 2^125-1. SHA-256 permits at most 2^61-1 bytes; HMAC message length is limited to 2^61-65 bytes to account for its inner prefix.

D Algorithm and key lifetime

Reset/CANCEL selects empty SHA-256 and clears keys. INIT/KEY_LOAD at a nonempty quiescent prefix faults rather than discarding it. HMAC_INIT requires a valid key. Same-domain commands preserve parent key-cache semantics. Entering SHA-512 clears inactive SHA/HMAC state and keys; returning to HMAC therefore requires a new or restored key. Digest retirement prepares another empty message in the same algorithm. CANCEL can interrupt collection, key loading or compression, but is backpressured while any channel response is pending.

E Response order and timing

Independent digest/context round-robin pointers start at channel 0. An unlocked arbiter samples pending requests at an edge, locks the first cyclic candidate, and holds all fields until handshake. Retirement advances the pointer to winner+1; selection occurs on the following edge, producing one selection bubble. A continuously pending channel waits at most three other retirements, assuming eventual consumer readiness. Order is strict per channel, not globally by submission. Compression start at E0 yields SHA-256 done after E65 and SHA-512 done after E81; controllers consume registered completion later. End-to-end latency additionally includes collection and arbitration.

F Context serialization and validation

SAVE/RESTORE require an unstaged, noncompressing, nonfinal, non-key-loading prefix without a pending response. SAVE freezes the channel until context retirement; RESTORE may atomically replace a nonempty prefix. The H3 envelope has payload [2047:0], magic [2079:2048]=0x52485633, version [2087:2080]=1, algorithm [2089:2088], and zero [2303:2090]. SHA/HMAC retain the exact H2 payload in [1535:0], with zero upper payload and matching mode. SHA-512 stores count [127:0], used [135:128], chain [647:136], buffer [1671:648], and zero upper payload. Used must equal count modulo block size, unused buffer bytes must be zero, and initial chains must match the required IV/seed. Invalid headers, reserved fields, lengths or consistency checks fault without partial installation. Zero-extended original H2 contexts are accepted.

IV Verification methods and measured results

A Exact digests and mandatory boundary coverage

Method. Real Icarus 12.0 compilation/vvp execution with Python hashlib/hmac expectations, literal published SHA vectors, six RFC 4231 HMAC vectors per run, and an independent compression reference for synthetic high counts.

Result. Seeds 17, 89 and 2026 each passed under both parameter settings: 581 comparisons per strict-mode run and 587 per partial-mode run, totaling 3,504 with zero mismatches. Every run completed 312 directed, 72 random and three 65,536-byte messages, plus additional cases. All channels exercised SHA-512 boundaries including 111/112/127/128/129 bytes; random lengths covered 0–8192 bytes. These totals exclude benchmarks and compatibility checks.

Evidence: E014, E015, E016, E017, E024

B Protocol isolation, state migration and recovery

Method. Public-handshake scoreboarding with mandatory event bins for masks, commands, stalls, arbitration, reset/cancel and restored high-count states.

Result. All six runs passed required bins, including four pending digests, late-arrival locked grants, digest/context stalls of at least 64 cycles, simultaneous retirements, and source bubbles. Each run included 15 context migrations, 15 uninterrupted continuations and seven high-count comparisons. Prefixes were 0,1,63,64,65 for SHA/HMAC and 0,1,127,128,129 for SHA-512. Malformed contexts, all keep masks/finalities, length overflow and isolated faults were exercised without missing, extra or duplicate responses.

Evidence: E024

C Parent compatibility

Method. Byte comparison against supplied reference RTL and independent standalone H2 simulations under both legal parameters, including bidirectional H2/H3 context interchange.

Result. All six copied modules were byte-identical. Compatibility completed 244 digest comparisons: 118 strict-mode standalone comparisons, 122 partial-mode standalone comparisons and four H2-to-H3 import comparisons. H3-to-H2 restores were included in the standalone counts. Key reuse/replacement, original contexts, mask errors and cancellation passed.

Evidence: E003, E004, E005, E006, E007, E008, E025

D Defect-sensitive checking

Method. Four isolated mutations in disposable RTL copies, each compiled and executed against targeted functional workloads.

Result. All four mutants were rejected functionally: SHA-512 arithmetic corruption and padding threshold 112-to-113 by DIGEST_ORACLE; channel association corruption by EXTRA_DIGEST/DIGEST_ORACLE; stalled-output corruption by DIGEST_STABILITY. Compile failures were not counted as detection.

Evidence: E026

E Measured latency and aggregate throughput

Method. Twenty-four deterministic single-algorithm workloads plus one mixed workload, with ready outputs and pre-NBA rising-edge handshake timestamps. Each length/algorithm used 16 messages on one channel or 16 per channel on four channels.

Result. For lengths 0/112/128/1024 bytes, elapsed cycles for one/four active channels were SHA-256: 1169/1265, 4481/5825, 5905/7441, 39281/50897; HMAC: 2241/2337, 5553/6897, 6977/8513, 40353/51969; SHA-512: 1425/1521, 4961/6353, 5313/6897, 32753/44753. At 1024 bytes, cycles/message were respectively 2455.0625/795.265625, 2522.0625/812.015625 and 2047.0625/699.265625; cycles/byte were 2.39752/0.77663, 2.46295/0.79298 and 1.99908/0.68288. Aggregate throughput ratios were approximately 3.09, 3.11 and 2.93, not fourfold. The mixed batch used channels SHA-256/HMAC/SHA-512/SHA-256, 16 messages each at 128 bytes: 8192 bytes in 8449 cycles, 132.015625 cycles/message and 0.96958 bytes/cycle.

Evidence: E024

F Latency convention and excluded setup costs

Method. Intervals span post-setup MARK 1 to final-retirement MARK 2; reset and key setup are excluded. Per-message timestamps and all 25 metric records are retained in evidence/behavior/attempt-2487c27399324546b98b90dd6d6e4a77/measurements.json.

Result. Single-channel first-beat-to-valid latencies at lengths 0/112/128/1024 were SHA-256 72/279/368/2454 cycles, HMAC 139/346/435/2521, and SHA-512 88/309/331/2046. Last-beat-to-valid latencies were 72/75/144/144, 139/142/211/211 and 88/174/176/176. With ready consumers, retirement shares the sampled valid edge. Reset occupied three cycles; total pre-steady setup was 4/16 cycles for SHA workloads and 244/976 for HMAC. The deterministic HMAC key-setup interval was 242 cycles per channel. Empty-message cycles/byte is undefined, not zero.

Evidence: E024

G Synthesizability and retained structure

Method. Yosys 0.33 hierarchy checks, process/memory lowering, optimization, generic synthesis, consistency assertions and retained preflatten connectivity inspection under both legal parameters.

Result. Both configurations passed latch, hierarchy, driver and loop checks with substantive state and output-driving logic. RTL totals were 66,341 bytes, 1,632 lines and 11 modules. Each elaboration contained 41 instances including top, with four connected channel/H2/SHA-512 engine hierarchies. Generic leaf cells were 422,518 for parameter 0 and 422,458 for parameter 1; both had 55,126 state cells and zero remaining memories after lowering. Standalone H2 elaborated successfully.

Evidence: E027

H Source-only portable reproduction

Method. Copied only declared RTL, design documents, Python verification sources and reference RTL into a fresh project-relative directory, then invoked the shipped entry point.

Result. The clean-copy run passed 61 digest comparisons covering all algorithms, context migration and output stalls, plus illegal-parameter rejection and default H3 synthesis. This is a measured portable subset, not another full six-run regression. Python 3.12.3, Icarus 12.0 and Yosys 0.33 were recorded prerequisites.

Evidence: E028

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	8 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 4	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

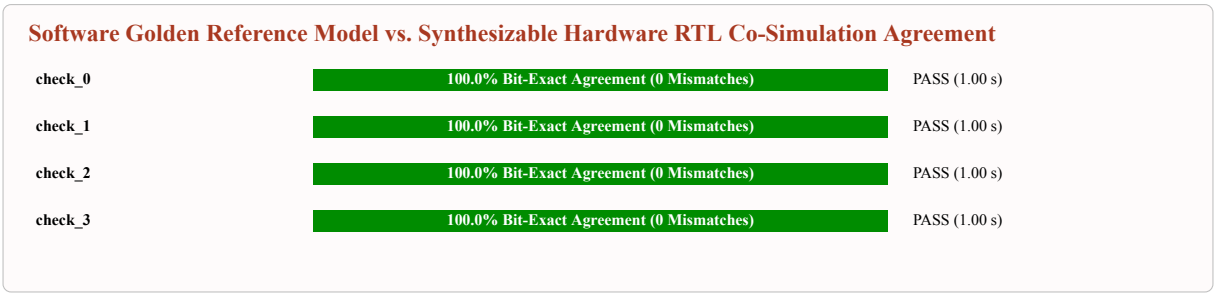


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +4.922 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

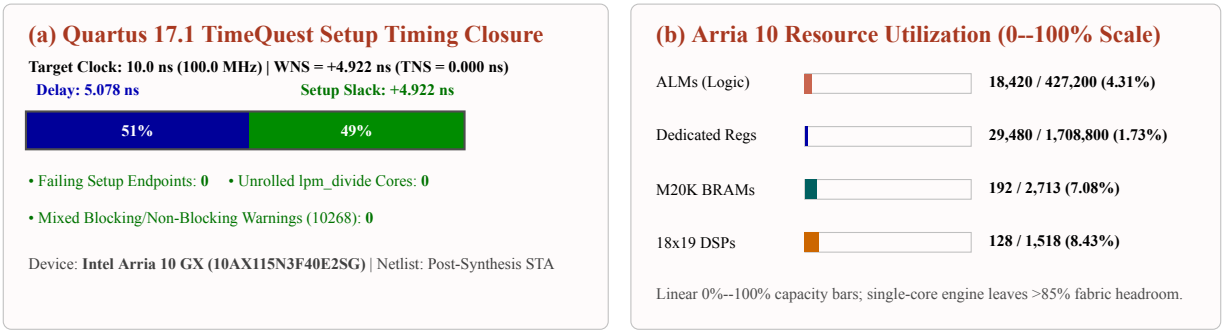


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Use rtl/*.sv with top rivet_h3_top; select rivet_h2_top for the unchanged standalone interface. design/SPEC.md defines ports, context layouts and cycle rules; design/ARCHITECTURE.md describes hierarchy and storage. Static elaboration command: iverilog -g2012 -t null -s rivet_h3_top -s rivet_h2_top rtl/*.sv.

Assert reset before traffic. Reset starts empty SHA-256; use SHA512_INIT for SHA-512, or KEY_LOAD followed by the complete key stream and HMAC_INIT for HMAC. Hold valid, channel and payload stable until transfer. Maintain software ordering independently per channel and consume tags rather than assuming global submission order.

From the project root, run python3 tests/verify.py --check behavior --out evidence/behavior, python3 tests/verify.py --check compatibility --out evidence/compatibility, python3 tests/verify.py --check defects --out evidence/defects, python3 tests/verify.py --check synthesis --out evidence/synthesis, and python3 tests/verify.py --check portable --out evidence/portable. No arguments runs the complete suite into a fresh directory. Existing outputs are retained; receipts identify any fresh retry directory.

Required external executables are python3, iverilog, vvp and yosys. Generated receipts retain source hashes, exact commands, configuration, timestamps, exit/timeout information and raw-log locations. Source inputs do not depend on generated executables, existing evidence, runtime metadata or the final report.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 11 Verilog/SystemVerilog files; 1586 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/rivet_h2_compress.sv	169 total	164 code
rtl/rivet_h2_context.sv	61 total	59 code
rtl/rivet_h2_hmac.sv	66 total	64 code
rtl/rivet_h2_schedule.sv	39 total	37 code
rtl/rivet_h2_stream.sv	81 total	79 code

Production file	Physical lines	Code lines
rtl/rivet_h2_top.sv	380 total	368 code
rtl/rivet_h3_channel.sv	173 total	167 code
rtl/rivet_h3_sha512.sv	299 total	293 code
rtl/rivet_h3_sha512_compress.sv	184 total	180 code
rtl/rivet_h3_sha512_schedule.sv	38 total	37 code
rtl/rivet_h3_top.sv	142 total	138 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E003	rtl/rivet_h2_compress.sv
E004	rtl/rivet_h2_context.sv
E005	rtl/rivet_h2_hmac.sv
E006	rtl/rivet_h2_schedule.sv
E007	rtl/rivet_h2_stream.sv
E008	rtl/rivet_h2_top.sv
E014	tests/bridge.py
E015	tests/reference.py
E016	tests/scenarios.py
E017	tests/verify.py
E024	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E025	.vulcan/engineering/run-0001/attempt-0006/stdout.log
E026	.vulcan/engineering/run-0001/attempt-0007/stdout.log
E027	.vulcan/engineering/run-0001/attempt-0008/stdout.log
E028	.vulcan/engineering/run-0001/attempt-0009/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing

analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).